

# HOT TECHNOLOGIES IN 2003

**T**hree years into the millennium, and we stand on the verge of yet another persuasive wave of technology as it entwines itself into the very fabric of our lives. What was once considered high technology is now accessible to a vast cross-section of users, ranging from teenagers to grandparents, as much as it is to technologists and scientists. We are used to an infusion of gadgets that have made themselves indispensable—be it cellular phones, ATM cards or ready access to the Internet. Let's take a look at some of the technologies that are going to be instrumental in furthering this acceptance of technology in our day-to-day lives. **d**

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## The wire-free network

**W**hile high-speed networks draw on all the advantages of your office Ethernet, high-bandwidth wireless networks spell untold levels of convenience and portability. With 802.11b (WiFi) networks already in place in many parts of the world, and present in numerous home and office networking equipment, there's no looking back. The 54 Mbps bandwidth of such networks provides adequate room for data-hungry applications such as streaming audio and video and high-volume file downloads. Also, the capability of providing up to eight discrete channels in the specification lets the network handle different types of data streams. While there were speed breakers in the shape of slow acceptance for the 802.11a standard and compatibility with other wireless standards, today, both the 802.11a and 802.11b standards can be merged into a single device. Another hybrid variant of the 802.11x standard is the 802.11g, which boasts of bridging the gap between the 802.11a and 802.11b standard, enabling them to talk to each other. On the horizon are other exciting wireless standards such as Ultra Wide Band (UWB) wireless communication. With its significantly higher levels of security, UWB is likely to gain wider recognition and acceptance in the forthcoming year.